

Assignment 1: Visualization in the World

DATA 3302: Data Visualization, Fall 2026

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I Assignment Overview

From the New York Times to fitness trackers and COVID-19 dashboards to Wordle shares, the world is awash in data and visualization. In this assignment, you will reflect on and take stock of the visualizations that pervade your own life, from the practical and functional to the whimsical and fun. You'll then analyze and critique these visualizations based on your current design knowledge.

II Part 1: Collection and Curation

In Part 1, take screenshots, photos, and/or videos of visualizations that you encounter in your personal, professional, and academic lives. The goal with this part is to reflect on the pervasiveness of visualizations in your life, observe their use and purpose, and finally analyze them. While you will certainly include some examples of “standard” visualizations like barcharts and the like, the best assignments will be able to also construe “data visualization” broadly. Really think about what are some different ways in which data is presented to you graphically for the purpose of sensemaking?

Collect a minimum of ten visualizations and organize them in a PDF slideshow (see Deliverables). For each visualization, enter a brief description and how it relates to your life. Make sure to include a citation to the source of the visualization if possible.

The examples should be grouped, ordered, or otherwise organized by some sensible approach that is up to you: you could organize them by chart type, by domain (health, schoolwork, finances, fun, etc.), by communicative purpose (to inform, to persuade, to explore), etc.

III Part 2: Critique

From your collection, **select three visualizations to critique and analyze**. For each of the three, address the following questions:

1. What is the goal of the visualization? What do you learn from it?
2. How is the data represented? How do the various graphical marks and visual elements represent the data being shown? This should be directly annotated.
3. What are some strengths and weaknesses of the visualization? In what ways does it do a good job? How can it be improved?

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4. What is the historical or cultural context from which this design comes from? Are there any important similarities between this visualization and other earlier visualizations that it is drawing from?
5. What are the different **user tasks**¹ that the visualization is designed to support?

IV Part 3: Reflection

Finally you must include a short, 1-2 paragraph learning reflection. You should address questions such as “In this assignment, I learned about... I had not previously considered... I was surprised that...”

V Useful Resources

To potentially aid you in finding, analyzing and critiquing the visualizations in your collection, here are some background readings/videos on the value of visualization, graphical composition, and the relationship between perception and visualization.

- Munzner. Intro: [Why visualize?](#) Ch. 1 of Visualization Analysis & Design.
- Munzner. [Marks and channels](#). Ch. 5 of Visualization Analysis & Design.
- Card, Mackinlay, Shneiderman. [Using Vision to Think](#). Ch 1. in Readings in Information Visualization.
- Heer, Bostock, and Ogievetsky. [A Tour Through the Visualization Zoo](#).
- Stone. [Choosing Colors for Data Visualization](#).
- [Information is Beautiful Award Winners](#)

VI Deliverables and grading

For this assignment there will be both a submitted file as well as an in-person presentation.

Gradescope submission

For for deliverable you must submit to gradescope a PDF slideshow including

1. A title intro slide
2. A slide describing how you organized your collection
3. Your collection in Part 1 where there is a slide for each visualization including:
 1. An image of the visualization
 2. Description of what it is and how you found it
 3. Source/link
4. Your critiques for Part 2 which for each chosen visualization must include a slide each detailing (briefly and in summary):
 1. The annotated visualization showing and describing the different visual elements and how they are mapped to from data
 2. A description of the context and goals of the visualization, what it is trying to say and to whom?
 3. Identification of the different user tasks that the visualization is designed to support.

¹Based on Munzner’s typology from lecture

4. An analysis of what elements do a good or poor job in communicating the goal, and potentially what could be done differently.
5. A slide including your reflection on the assignment.

Presentation

After you have submitted your presentation slides PDF, you will be able to actually present your findings to the rest of the class during scheduled lab time during Week 3. If you are not finished by the time of the scheduled presentations, or if you want to revise and resubmit your assignment you can re-present one-on-one during office hours or during extra time later during lab.

Grade

You will be assessed based on:

1. Inclusion of all of the required components of the assignment.
2. Clear presentation detailing meaningful engagement with the chosen visualizations.
3. Your ability to respond to questions about the visualizations and your analysis beyond what is written directly on the slide.

VII Acknowledgement

This assignment is based on assignments by [Jon Froehlich](#) and [Ayaan Kazernouni](#).